

Materials and Methods

Structure-corrected two-locus association for intrinsic incompatibilities between unlinked LD-reduced units

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Draft prepared from the current /doc pipeline documentation

This is a self-contained working draft for integration into the full Supplementary Materials, describing the intrinsic (Dobzhansky–Muller) arm that complements the ancestry-sorting and climate analyses documented separately. Blue boxes identify items that remain open; in particular the neutral, recombination-matched null (Section S7) is still required to convert the descriptive leads into excess-over-neutral statements.

S1 Scope and analytical principles

These methods describe a screen for pair-specific, two-locus incompatibilities between *unlinked* LD-reduced units, complementary to the per-locus ancestry-sorting analyses (which measure the *direction* and extent of sorting) and to the climate analyses (which test the extrinsic axis). The same LD-reduced units, expected multi-locus genotype (eMLG) consensus dosages, and parental gate are reused; the methods for their construction are given in the companion document.

An intrinsic incompatibility can leave three traces in replicate hybrids, and separating them organises the analysis. (1) *Unidirectional* sorting, in which one parental combination is fitter, drives the same ancestry toward fixation repeatedly; this is the per-locus signal of the sorting analyses and is, by construction, invisible to any two-locus test, because linkage disequilibrium is bounded by the allele frequencies and vanishes as either locus approaches fixation. (2) *Bidirectional* sorting, expected under an equal-fitness incompatibility, fixes opposite compatible combinations in different replicates; its single-locus parallelism test is null by construction, and its signal lives at the pair level. (3) *Coupling maintained within populations* is elevated two-locus LD inside demes, which is badly confounded by cryptic relatedness among nestmates. A two-locus screen must therefore condition on both the genome-wide ancestry gradient and the colony structure before any residual association can be read as pair-specific.

Three principles guided the analysis, consistent with the companion methods. First, the LD-reduced units are treated as complexity-reduced analytical units, not as strictly independent tests; residual dependence is controlled or annotated where it matters. Second, a variable used as a predictor was not also used to select the analysed data. Third, the twenty hybrid populations are not assumed to be fully independent replicates; analyses using among-population variation either condition on it or are interpreted descriptively, and the final intrinsic call is deferred to a neutral simulation.

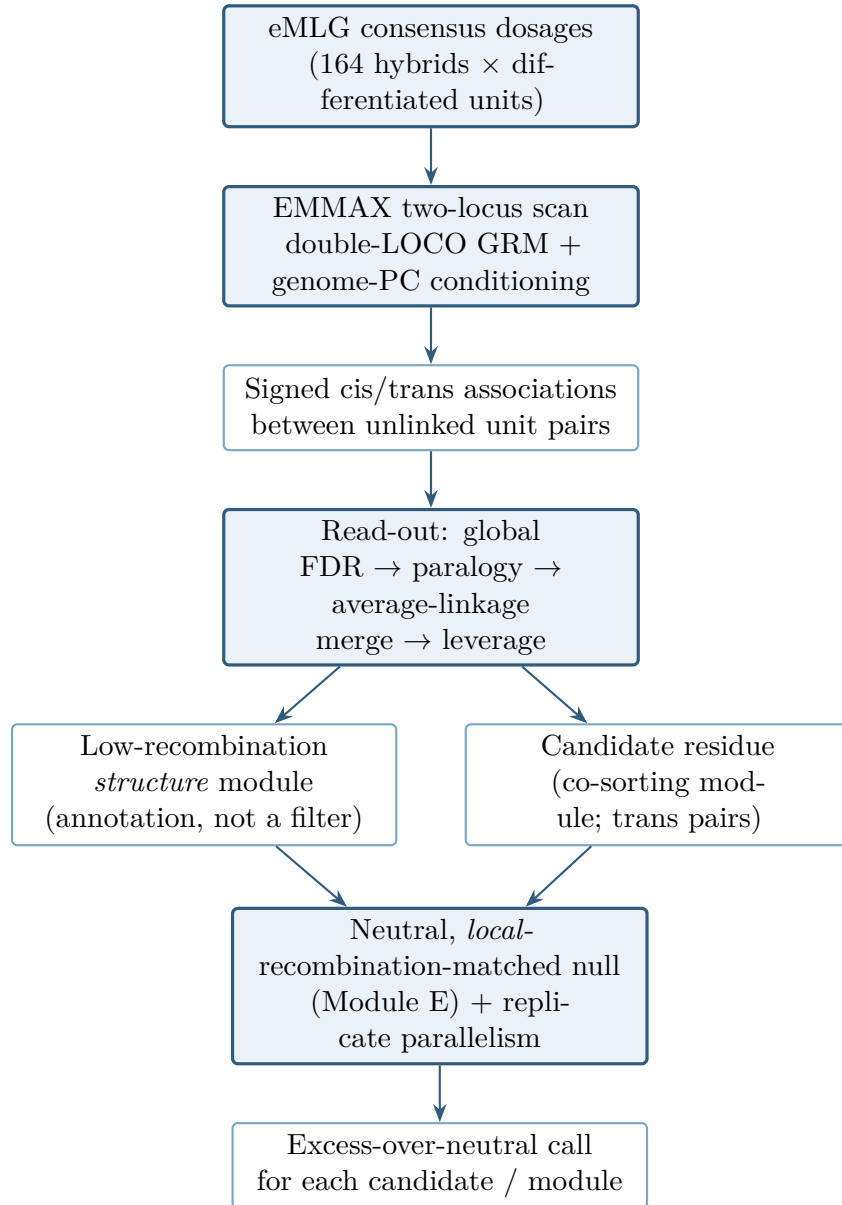


Figure 1: **Analytical workflow for the intrinsic arm.** A single structure-corrected scan is read out by one reproducible procedure into a low-recombination co-ancestry module and a short candidate residue. Low recombination is annotated, never used to discard loci; every unit is carried into the neutral, local-recombination-matched null that provides the final call.

S2 Units, orientation, and the unlinked definition

The analytical unit is the eMLG consensus dosage of an LD cluster (continuous, on an *F. aquilonia*-oriented 0–2 scale; no hard-calling), restricted to the parent-differentiated units used as informative markers throughout. Consensus orientation and the parental gate (pooled parental MAF ≥ 0.15) follow the companion methods, so that a unit’s dosage and its diagnostic index (DI) are on consistent scales.

Two units were treated as *unlinked* when they lay on different chromosomes, or on the same chromosome more than $\text{LINK}_{\text{cM}} = 10$ cM apart on the genetic map. Distance was taken on the map rather than in base pairs because admixture LD decays with the recombination fraction, not with physical distance: at the recombination rates of this genome, 10 cM corresponds to $\sim 99\%$ decay of admixture LD over the ~ 50 generations since contact. Per-unit map positions were the median member-marker position interpolated from the recombination map. This is the same threshold used to define the LD-reduced units, so “unlinked” pairs are those the clustering already treats as distinct regions.

S3 Structure-corrected two-locus association (EMMAX)

S3.1 Rationale and model

To ask whether two units are associated *beyond* what shared ancestry and relatedness predict, we treat one unit’s consensus dosage as a quantitative trait and test its association against every other differentiated unit in a linear mixed model,

$$y_i = \mu + \beta x_j + u + \varepsilon, \quad u \sim N(0, \sigma_g^2 K), \quad \varepsilon \sim N(0, \sigma_e^2 I), \quad (1)$$

where y_i is the focal unit’s dosage across the 164 individuals, x_j a candidate partner, and K a genomic-relationship matrix (EMMAX; Li, Kempainen, Rastas & Merilä). This is a genome scan in which the “phenotype” is one unit and the “markers” are all the others; variance components are estimated once under the null and reused for every partner.

S3.2 Why one random effect handles both confounds

The leading eigenvectors of K are the ancestry/structure principal components, so conditioning on K removes the genome-wide ancestry gradient at the individual level; and because K also encodes sibling relatedness among nestmates, the Wahlund inflation of within-population LD is regressed out in the *same* term. Three choices are load-bearing:

1. **Basis of K .** K was built from the ancestry-*informative* (differentiated) units, not a neutral, LD-pruned background. The confound to be absorbed is the *selected* ancestry-sorting axis carried by those units; a neutral basis dilutes that eigenvector and under-corrects, inflating the genomic control factor to $\lambda \sim 1.2$ and leaking the ancestry gradient back as spurious positive associations.
2. **Double leave-one-chromosome-out.** Because the scan is symmetric, both the focal and the partner chromosome are dropped from K , so neither tested unit’s own region deflates its association. This keeps the test calibrated ($\lambda \approx 0.98$ – 1.05 ; Fig. 2).
3. **Genome-PC conditioning.** Each focal trait was residualised on the top ten genome-wide principal components before the mixed model, because K cannot remove a *secondary*

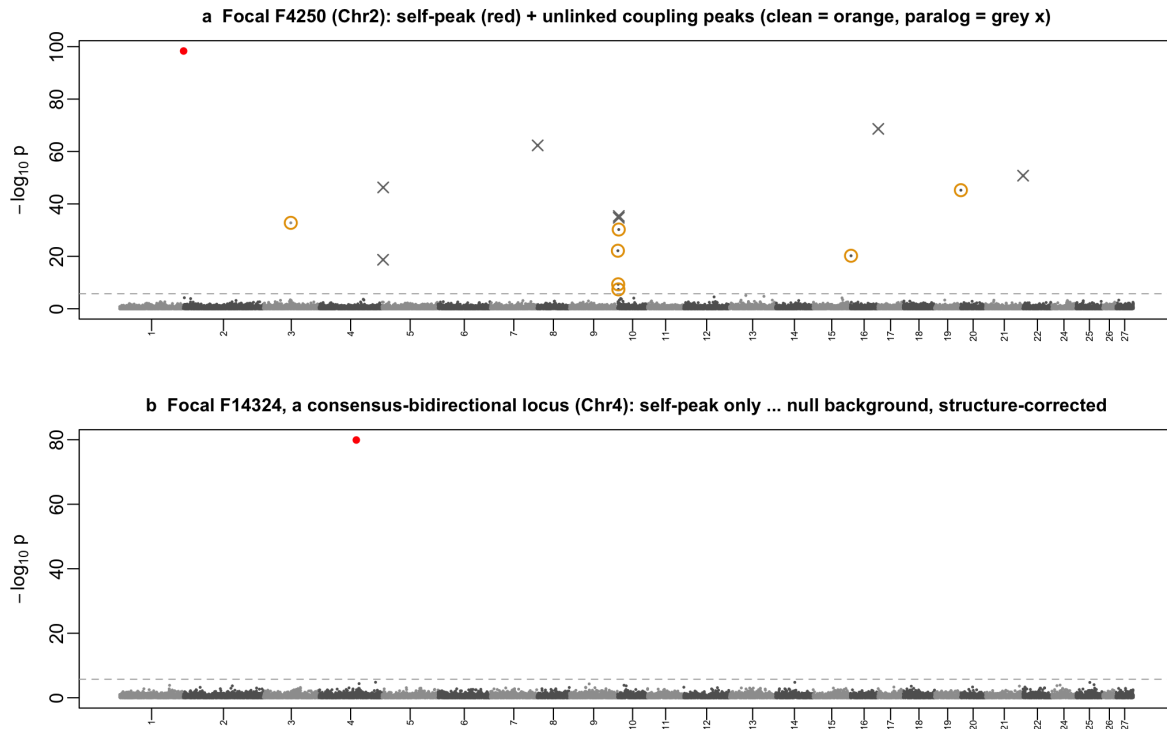


Figure 2: **Calibration and read-out of the structure-corrected scan.** Diagnostic summary of the double-LOCO, genome-PC-conditioned EMMAX scan: quantile–quantile calibration of the association statistic ($\lambda \approx 0.98$ – 1.05), an example focal Manhattan with its self-locus and linked block removed, and the signed (cis/trans) genome-wide hits. The scan is calibrated and its strongest associations are cross-chromosome, motivating the paralogy filter of Section S4.

structure axis concentrated on the very chromosomes double-LOCO drops. After this step the random-focal control returns essentially no significant partner, while calibration holds.

Each association is signed by its structure-corrected generalised-least-squares coefficient (coupling, +; repulsion, –), recovering both cis and trans hits. A first, targeted run scanned 51 focal units (bidirectional candidates, extreme among-population-covariance units, and controls including a random-focal set for calibration).

Open item. An unbiased genome-wide focal scan (or a larger random focal sample) is required to report an honest genome-wide hit rate rather than the enrichment of a targeted focal set; a full all-versus-all scan ($\sim 10^9$ tests) is unnecessary and the random control fixes the scale. This is scheduled with the Module E run.

S4 Cross-chromosome paralogy, and why distance cannot remove it

The genetic-distance rule cannot catch cross-chromosome duplicates: two units on different chromosomes are at infinite recombinational distance yet can be genotype-near-identical when the same reads map to two assembly locations. These pairs are the loudest inter-chromosomal “LD” and are technical rather than epistatic. Both the among-population and the structure-corrected read-outs therefore apply a signal-based filter.

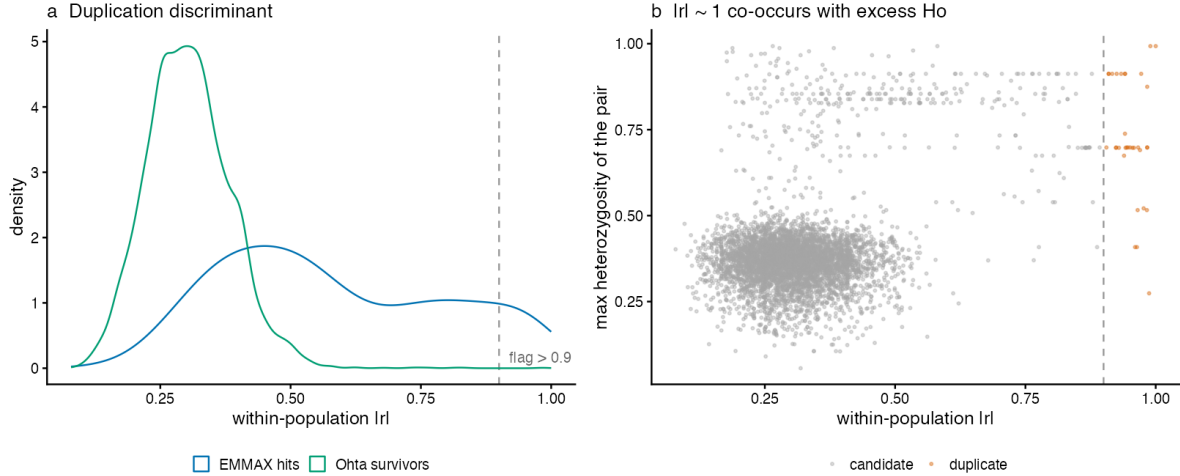


Figure 3: **The paralogy discriminant.** Distribution of median within-population $|r|$ for unlinked pairs, with the 0.9 flag; the corroborating rise in genotype concordance and per-unit excess heterozygosity toward $|r| = 1$; and the separation of flagged (technical) from retained (candidate) pairs. Distance filtering is orthogonal to this signal-based filter, so both are applied.

The discriminant is the individual-level *within-population* correlation of the two units' consensus dosages, taken as the median over populations. Within-population correlation removes the among-population (ancestry-axis) component, which is not what paralogy is about; real long-range coupling in these data is moderate ($|r| \lesssim 0.6$), whereas a duplication sits at $|r| \approx 1$ (a flipped duplicate gives $r \approx -1$, which $|r|$ still catches). A pair was flagged as paralogous when the median within-population $|r|$ exceeded $\text{PARALOGY}_r = 0.9$. Genotype concordance and per-unit excess observed heterozygosity (a duplicated region inflates H_o) are stored alongside as corroboration (Fig. 3). The flag is treated as a paralogy *candidate* rather than a proof, because an extremely strong genuine within-deme incompatibility could also approach $|r| = 1$; the grey zone 0.6–0.9 is retained as clean, conservatively.

S5 A single reproducible read-out of the scan

The scan was read out by one procedure rather than per-focal thresholds, so that the structural and pair-specific components separate under a single criterion.

1. **Global false-discovery control.** All unlinked (focal, partner) tests (1.38×10^6) were pooled and controlled at a global Benjamini–Hochberg $q < Q_{\text{FDR}} = 0.01$ — one whole-experiment error rate rather than 51 independent per-focal corrections.
2. **Paralogy filter** (Section S4).
3. **Third-level, context-only merge.** For this network alone (not the canonical clustering), same-chromosome units within LINK_{cM} that remain correlated were collapsed into meta-nodes by *average-linkage* clustering on $1 - |r|$, cut at $|r| > \text{MERGE}_r = 0.5$, with pairs beyond LINK_{cM} given an un-mergeable distance. Average linkage requires the *whole* merged group to be mutually correlated; single linkage (connected components) chains uncorrelated ends and is not used. This prevents one low-recombination region from reading as many interacting loci.
4. **Single-population leverage filter.** An edge was retained only if its among-population

allele-frequency correlation survived leave-one-population-out ($\min_k |r_{-k}| \geq \text{LEV}_{\text{LOO}} = 0.3$). Because a rare-allele edge that one deme drives fails this test, it subsumes any near-fixed / low-MAF cutoff, and no separate MAF filter is applied on the MAF-gated data.

Cross-chromosome correlation *modules* were then defined by average-linkage clustering of the meta-node consensuses *across* chromosomes, cut at $|r| > \text{MODULE}_r = 0.4$. Unlike the within-chromosome merge, this groups co-varying regions genome-wide, so each module is one coherent co-ancestry block used for display and for the single-module genotype views.

S6 Recombination annotation, not a filter

Each meta-node was annotated by its recombination percentile against the differentiated background; units below the tenth percentile ($\text{STRUCT}_{\text{pct}} = 0.10$; centromeric and pericentromeric) were labelled *structure*. This label is used only as an annotation and *never* as a filter, for a specific reason. Low recombination is not evidence of neutrality: it is where coadapted and epistatic complexes are expected to accumulate and persist, since recombination suppression is itself a hallmark of coadaptation. For an unlinked pair the between-region recombination fraction is 0.5 regardless of local rate, so low local recombination does not “hold” cross-chromosome LD; it makes each region a clean, low-noise ancestry block, i.e. the highest-signal marker of whatever residual covariation remains after genome-PC conditioning — finer neutral structure *or* selection. Every unit, low-recombination included, is therefore carried into the null of Section S7 and judged against a baseline matched to its own local recombination.

S7 Neutral, recombination-matched null

Every filter above removes structure *aligned* with what it conditions on; none can say whether a residual pair, or a low-recombination module, exceeds what neutral, recombination-matched drift would produce in twenty replicate populations of this size and design. That is the role of the neutral simulation (Module E), and two features are required.

- **Local-recombination matching.** Neutral admixture is simulated on the *real* recombination map, with matched founder ancestries, generations, and sampling, and the identical scan and read-out (Sections S3–S5) are applied to the simulated genotypes. A pair or module is a candidate only if its observed LD exceeds the neutral null *at its own local recombination rate*, so a low-recombination pair is compared against a legitimately high baseline and can still surface.
- **Replicate-population parallelism.** Neutral relic co-ancestry drifts in random directions across the twenty populations, whereas epistatic selection maintains the same combination reproducibly; a module whose direction is parallel across many populations, or that exceeds the general low-recombination background, is corroborated as selection independently of the simulation.

Open item. Module E is not yet run. Decide whether local matching is achieved by simulating directly on the real per-chromosome map (preferred; no binning artefacts) or by binning observed pairs on local cM/Mb and comparing to rate-matched simulated pairs.

S8 Statistical interpretation

The analyses are descriptive and conditional on the sampled populations, quality gates, and analysis definitions. Structure correction removes covariation aligned with the ancestry axis and colony relatedness; the recombination annotation flags where the neutral LD baseline is high; and the genotype-level PCA characterises the structure module without proving its neutrality (parallel or stabilising selection can also lower among-population differentiation). No pair is called an incompatibility on the basis of the association scan alone. The excess-over-neutral statements are made only against the recombination-matched null, and individual candidates then require independent biological scrutiny.

Table 1: Principal parameter settings for the intrinsic (two-locus) arm.

Analysis	Parameter	Value
Units	Analytical unit	eMLG consensus dosage (differentiated)
	Parental MAF gate	≥ 0.15
Unlinked	Genetic-distance rule	different chr, or same chr > 10 cM
EMMAX	Model	LMM, one unit as trait vs all others
	Genomic-relationship matrix K	VanRaden, differentiated units
	Leave-one-out	double (focal + partner chromosome)
	Genome-PC conditioning	top 10 PCs residualised from the focal
	Sign	structure-corrected GLS coefficient
	Calibration	$\lambda \approx 0.98\text{--}1.05$
Paralogy	Discriminant	median within-population $ r $
	Flag threshold	> 0.9
Read-out	Multiple testing	global Benjamini–Hochberg $q < 0.01$
	Third-level merge	average linkage, $ r > 0.5, \leq 10$ cM
	Leverage	leave-one-population-out $ r \geq 0.3$
	Modules	average linkage across chr, $ r > 0.4$
Annotation	Structure label (not a filter)	recombination percentile < 0.10

Analysis	Parameter	Value
Null (Module E)	Matching	local recombination (real map)
	Corroboration	replicate-population parallelism

References

- [1] Kang HM, Sul JH, Service SK, Zaitlen NA, Kong S-Y, Freimer NB, Sabatti C, Eskin E. Variance component model to account for sample structure in genome-wide association studies. *Nature Genetics*. 2010;42:348–354.
- [2] Nouhaud P, Martin SH, Portinha B, Sousa VC, Kulmuni J. Rapid and predictable genome evolution across three hybrid ant populations. *PLoS Biology*. 2022;20:e3001914. doi:10.1371/journal.pbio.3001914.